

**Decoding the Conformational Landscape of Ribonucleotide
Reductase: Ensemble Modelling of Radical Transfer, Allosteric
Communication, and Dynamic Regulation Using BioEmu**

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Declaration

The work embodied in the thesis entitled “*Decoding the Conformational Landscape of Class Ib Ribonucleotide Reductase: Ensemble Modelling of Radical Transfer, Allosteric Communication, and Dynamic Regulation Using BioEmu*” was carried out by me under the supervision of Dr Shekhar Mande, Distinguished Professor, Savitribai Phule Pune University, in partial fulfilment of the requirements for the award of the M.Sc. degree in Bioinformatics. The work presented in this thesis is original and has not been submitted, either in part or in full, to this or any other University or Institution for the award of any Degree or Diploma.

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Table of Contents

LIST OF ABBREVIATIONS	2
LIST OF TABLES	3
LIST OF FIGURES	4
ABSTRACT	5
1. INTRODUCTION	6
2. OBJECTIVE	8
3. REVIEW OF LITERATURE	9
3.1 Structural and Mechanistic Foundations of Ribonucleotide Reductase	
3.2 Conformational Heterogeneity as a Functional Requirement for RNR Catalysis	
3.3 Class Ib RNR Architecture and the NrdI-Dependent Radical Assembly Mechanism	
3.4 BioEmu-1: A Generative Deep Learning Framework for Protein Conformational Ensemble Generation	
4. METHODOLOGY	14
4.1 Molecular Dynamics Trajectory Collection and Preparation	
4.2 Structural Alignment and Global Motion Removal	
4.3 Structural Stability Assessment: RMSD Analysis	
4.4 Residue-Level Flexibility Analysis: RMSF	
4.5 Correlated Motion Analysis: Dynamical Cross-Correlation Matrix	
4.6 Residue Interaction Network Construction and Analysis	
4.7 Hub Identification and Centrality Analysis	
4.8 Quality Control and Reproducibility	
5. RESULTS	18
5.1 Conformational Dynamics of NrdB (PDB ID: 3EE4)	
5.2 Conformational Dynamics of NrdI (PDB ID: 8J4V)	
5.3 NrdF2:NrdI Complex Dynamics (PDB ID: 8J4X)	
6. DISCUSSION	33
6.1 BioEmu Captures Functionally Relevant Conformational Landscapes	
6.2 The 50s-Loop of NrdI is the Principal Dynamic Regulatory Element	
6.3 Residue 54 and PRO47 Define a Communication Bridge in NrdI	
6.4 NrdF2 Maintains a Stable Catalytic Core with Localised Communication Hotspots	
6.5 Asymmetric Functional Organisation of the NrdF2:NrdI Complex	
7. CONCLUSION AND FUTURE SCOPE	37
8. REFERENCES	39
9. APPENDICES	41

Abbreviations

Ribonucleotide reductase	RNR
Proton-coupled electron transfer	PCET
X-ray free electron laser	XFEL
BioEmu	Biological Emulator (BioEmu-1)
MD	Molecular Dynamics
RMSD	Root-Mean-Square Deviation
RMSF	Root-Mean-Square Fluctuation
DCCM	Dynamical Cross-Correlation Matrix

List of Tables

Table 1. Summary of RMSD statistics for the NrdB (3EE4) BioEmu ensemble.	19
Table 2. Key flexibility statistics for the NrdB (3EE4) ensemble.	20
Table 3. Hub residue distribution across structural regions of NrdB (3EE4).	22
Table 4. Inter-residue contact statistics for the NrdB BioEmu ensemble versus the 3EE4 crystal structure.	23
Table 5. RMSD statistics for the NrdI (8J4V) BioEmu ensemble.	24
Table 6. Key flexibility features of the NrdI (8J4V) ensemble.	26
Table 7. Key hub residues identified by betweenness centrality analysis in NrdI (8J4V)	27
Table 8. Structural flexibility summary for NrdF2 (8J4X complex).	28
Table 9. Principal communication hub clusters identified in NrdF2 (8J4X complex)	30
Table 10. Hub residue centrality values for NrdI in the 8J4X complex.	32

List of Figures

Figure 1. The backbone RMSD of every sampled conformation relative to the medoid reference frame (frame 37)

Figure 1. The RMSF analysis, which revealed a heterogeneous distribution of conformational mobility across the NrdB sequence

Figure 2. Structural visualisation of NrdB (PDB: 3EE4) mapped with dynamic and network features.

Figure 3. A Dynamic cross-correlation matrix (DCCM) contact map of NrdB

Figure 5. BioEmu contact probability map and contact difference analysis relative to the crystal structure.

Figure 6. RMSD analysis across the ensemble for NrdI

Figure 7. The RMSF analysis, which revealed a heterogeneous distribution of conformational mobility across the NrdI sequence

Figure 8. Structural visualisation of NrdI (PDB: 8J4V) showing the bipartite dynamic organisation.

Figure 9. DCCM and Network Analysis of NrdI.

Figure 10. RMSD Analysis of NRDF2 (PDB ID:8J4X)

Figure 11. RMSF Analysis of NrdI

Figure 12. DCCM and Network Analysis of NRDF2(8J4X)

Figure 13. Structural overview of the NrdF2:NrdI complex (PDB: 8J4X). NrdF2 subunits are shown in grey and NrdI subunits in light blue.

Figure 14. RMSD Analysis of NRDF2

Figure 15. RMSF Analysis of NRDF2(PDB ID :8J4X)

Figure 16. DCCM and Network Analysis of NRDI from the complex

Abstract

Ribonucleotide reductase (RNR) is the sole enzyme responsible for the de novo synthesis of deoxyribonucleotides, making it indispensable for DNA replication, repair, and cellular proliferation. Despite decades of structural investigation, the transient conformational states governing radical transfer, allosteric regulation, and inter-subunit communication in the Class Ib RNR system have remained largely inaccessible to experimental characterisation alone. This study employed BioEmu, a generative diffusion model trained on large-scale molecular dynamics simulations, to generate thermodynamically grounded conformational ensembles for three components of the Class Ia **NrdB** (*Mycobacterium tuberculosis*, PDB: 3EE4), else **NrdI** (*Mycobacterium thermoresistibile*, PDB: 8J4V), and the **NrdF2:NrdI** complex (*Mycobacterium tuberculosis*, PDB: 8J4X) associated with Class Ib. Ensemble analysis using RMSF profiling, dynamical cross-correlation matrices, and residue interaction network centrality revealed a conserved dual dynamic organisation across all three proteins, comprising a structurally rigid catalytic core flanked by discrete regions of elevated conformational mobility. In **NrdB**, the ensemble preserved native contact topology while sampling transient expanded states, identifying a rigid core centred on **residue 183**, flexible segments at **residues 75–89 and 142–155**, and communication hubs concentrated within **residues 103–151**, consistent with its role as a radical-harboring scaffold. The 50s-loop of NrdI (**residues 48–63**) was identified as the principal dynamic regulatory element of the complex, with its intrinsic pre-sampled flexibility providing a thermodynamic basis for redox-coupled oxygen-tunnel gating. Network centrality analysis further identified PRO47 and residue 54 in NrdI, alongside hub clusters at residues 88–98 and 208–214 in NrdF2, as the primary nodes of intramolecular communication. These findings provide the first ensemble-level mechanistic description of conformational regulation in the mycobacterial Class Ib RNR system, and establish BioEmu-driven network analysis as a powerful and generalisable framework for investigating dynamic allostery in radical enzyme complexes.

Keywords: *Ribonucleotide reductase, conformational ensemble, protein dynamics, BioEmu, residue interaction network, allosteric communication*

INTRODUCTION

DNA replication is one of the most fundamental processes in all living organisms, yet it cannot start without the very building blocks that make up DNA itself. At the centre of these building blocks sits ribonucleotide reductase (RNR), the sole enzyme responsible for synthesising deoxyribonucleotides from their ribonucleotide precursors through a de novo biosynthetic pathway. Because no alternative route exists for this conversion, RNR occupies an irreplaceable position in cellular metabolism, underpinning DNA replication, genome repair, cell division, and immune function alike. Due to this central biological role, RNR has become an important therapeutic target, with several clinically approved anticancer and antiviral agents acting through inhibition of its activity.

Among the structurally characterised classes of RNR, the Class I enzymes, particularly the Class Ib subtype found in many aerobic bacteria, exhibit a remarkably sophisticated architecture. They comprise two functionally distinct subunits: the catalytic NrdE subunit, which houses the active site and multiple allosteric regulatory regions, and the radical-generating NrdF subunit, which coordinates a dinuclear metal cofactor and maintains a stable tyrosyl radical that is essential for initiating catalysis. A dedicated accessory flavoprotein, NrdI, further assists in assembling and maintaining this radical centre under aerobic conditions. The catalytic mechanism itself relies on a long-range proton-coupled electron transfer (PCET) cascade that propagates a radical species across more than 35 Å, through a conserved chain of amino acid residues, to reach the active site and drive substrate reduction (Stubbe et al., 2008).

What makes RNR particularly challenging to study is that its catalytic cycle is deeply intertwined with dynamic conformational changes across the enzyme complex. The functional $\alpha_2\beta_2$ assembly is not a rigid static structure but rather an ensemble of interconverting conformational states, driven by C-terminal tail movements, metal centre rearrangements, and nucleotide-dependent allosteric transitions. Many of these states are transient, existing on timescales that are difficult to capture through conventional experimental methods such as X-ray crystallography or cryo-electron microscopy alone. Understanding these dynamics is therefore not merely a structural curiosity it is essential for

understanding how radical transfer is regulated, how substrate selectivity is enforced, and how allosteric signals propagate through the enzyme.

To address this challenge, the present study uses BioEmu, a recently developed generative diffusion model from Microsoft Research that represents a conceptual departure from conventional structure prediction tools. Rather than inferring protein dynamics as a secondary consequence of static structural data, BioEmu is trained directly on large-scale molecular dynamics simulation trajectories that include over 200 milliseconds of molecular dynamics (MD) simulations across thousands of protein systems, making it one of the largest publicly available sequence-diverse MD datasets. Enabling it to generate thermodynamically grounded conformational ensembles that reflect the equilibrium behaviour of proteins. BioEmu's training framework also incorporated AlphaFold2-predicted structures for more than 100 million proteins from the AlphaFold Protein Structure Database, along with over 500,000 experimental measurements of protein thermodynamic stability. By integrating simulation-derived dynamics, structural predictions, and experimental stability data, BioEmu was able to learn thermodynamically relevant protein conformational landscapes and generate realistic equilibrium structural ensembles. By using BioEmu analysis and by benchmarking against available experimental data, this work aims to construct a physically realistic description of the conformational landscape of the Class Ib RNR system. In doing so, we seek to highlight the structural basis of radical transfer, allosteric regulation, and catalytic gating in an enzyme that has a mechanistic complexity that long resisted complete characterisation.

OBJECTIVES

- To investigate the structural dynamics and conformational flexibility of ribonucleotide reductase proteins using computational and trajectory-based analysis approaches.
- To identify functionally important flexible regions and communication hub residues involved in catalytic activity, protein stability, and intermolecular interactions.
- To understand the relationship between protein dynamics and the functional mechanism of ribonucleotide reductase through structure-guided bioinformatics analysis.

REVIEW OF LITERATURE

2.1 Structural and Mechanistic Foundations of Ribonucleotide Reductase

Ribonucleotide reductase (RNR) has been a subject of sustained biochemical and structural investigation for several decades, due to its indispensable role as the sole enzyme capable of catalysing the de novo conversion of ribonucleoside diphosphates into their corresponding deoxyribonucleoside diphosphates, the essential precursor building blocks required for DNA synthesis and repair (**Nordlund and Reichard, 2006**). This reaction occupies a uniquely critical position in cellular metabolism, because no alternative biosynthetic route exists for the production of deoxyribonucleotides. RNR functions as an obligate checkpoint governing the rate of DNA replication across all domains of life.

The foundational structural characterisation of RNR has been progressively refined through successive generations of crystallographic and biochemical analysis. Early crystallographic studies resolved the overall subunit architecture of Class I enzymes, establishing the dimeric organisation of both the large catalytic subunit (NrdE) and the smaller radical-containing subunit (NrdF), and describing the spatial arrangement of the active site, allosteric regulatory pockets, and the di-metal cofactor centre responsible for stable tyrosyl radical storage. Subsequent accumulation of crystal structures across all three known RNR classes. These crystal structures have provided deeper insight into both the mechanistic diversity and the evolutionary conservation of this enzyme family (Nordlund et al., 2006). Comparative structural analyses have revealed that, despite differences in cofactor chemistry between the classes, the core fold of the catalytic subunit and the geometry of the active site cysteine residues are remarkably preserved, highlighting the functional role of radical-based substrate reduction.

A particularly significant advance in understanding RNR structural dynamics came from high-resolution room-temperature crystallography of the Class Ia R2 subunit, obtained using X-ray free electron laser (XFEL) serial femtosecond crystallography. This approach, revealed that the tyrosyl radical state is associated with explicit conformational reorganisation within the R2 subunit. Changes that act to shield the radical from solvent exposure while simultaneously establishing a structural connection to the translocation pathway through which the radical signal is communicated to the catalytic R1 subunit (**Nordlund et al., 2006**).

This finding was conceptually important because it demonstrated that radical shielding is not a passive thermodynamic consequence of the di-metal centre chemistry, but rather an active process coupled to protein conformational rearrangement. The structural changes observed propagate outward to the protein surface at the R1-R2 interface, implicating conformational dynamics directly in the inter-subunit radical relay mechanism.

2.2 Conformational Heterogeneity as a Functional Requirement for RNR Catalysis

The recognition that enzyme catalysis is intrinsically coupled to conformational dynamics, rather than operating through a rigid, static active site. This has fundamentally reshaped the conceptual framework of enzymology. For RNR, this principle has been established with particular clarity through a series of high-resolution structural studies employing cryo-electron microscopy combined with deep-learning-based classification algorithms to resolve distinct conformational substates populated during active enzymatic turnover (**Xu et al., 2025**).

Xu et al. (2025) demonstrated that Class I RNR does not adopt a single dominant conformation during catalysis, but instead continuously samples a broad conformational space encompassing a wide range of structural motions spanning multiple length scales. Critically, this conformational heterogeneity is not randomly distributed; the enzyme exhibits a marked and reproducible asymmetry between the two halves of the catalytic complex across different phases of the turnover cycle. This asymmetry appears to serve a regulatory function, enabling the two active sites within the homodimer to operate in a coordinated but non-synchronous manner. A mechanism likely relevant to the fidelity of nucleotide pool maintenance. These observations establish conformational heterogeneity as a genuine and functional feature of RNR catalysis, not merely structural noise arising from thermal fluctuations or crystal-packings.

The effect of these findings extends beyond the description of individual structural states. The rates of inter-state transitions, the relative thermodynamic populations of distinct conformers, and the asymmetric distribution of these populations across the two halves of the catalytic cycle collectively determine the overall catalytic efficiency of RNR. As **Xu et al. (2025)** argue, a complete mechanistic description of RNR therefore requires not merely the

identification of static structural snapshots with high resolution, but also the quantification of conformational dynamics as an integral dimension of enzyme function. This perspective makes the study of RNR dynamics a necessity rather than just a structural curiosity.

2.3 Class Ib RNR Architecture and the NrdI-Dependent Radical Assembly Mechanism

The Class Ib RNR system, characterised by the NrdE catalytic subunit and the manganese-dependent NrdF radical subunit, represents a biochemically distinct variant of the Class I RNR family that has attracted increasing structural and mechanistic attention in recent years. Unlike the iron-dependent Class Ia system, Class Ib RNRs employ atmospheric molecular oxygen as the terminal oxidant for radical generation at the di-manganese active centre of NrdF, a process that requires the participation of a dedicated flavoprotein accessory factor, NrdI (**Srinivas et al., 2018**).

Within the Class Ib catalytic cycle, ribonucleoside diphosphate reduction proceeds within the NrdE subunit through the canonical cysteine thiyl radical mechanism shared with other Class I enzymes. However, the generation and maintenance of the essential tyrosyl radical at the NrdF di-manganese centre proceeds through a mechanistic pathway that is unique to this subclass. **Srinivas et al. (2018)** provided important mechanistic characterisation of this pathway, demonstrating that NrdI facilitates the superoxide (O₂⁻)-mediated assembly of a Mn intermediate at the NrdF metal centre, which subsequently oxidises the conserved tyrosine residue to generate the stable tyrosyl radical (Tyr) required for catalytic competence. This oxygen-dependent radical assembly mechanism has been directly observed experimentally, providing a biochemical explanation for the aerobic growth niche occupied by Class Ib RNR-containing organisms.

Despite this mechanistic progress, several important questions surrounding Class Ib RNR remain unanswered. The dynamic assembly of the tri-protein NrdE-NrdF-NrdI complex, the conformational changes within NrdF that couple metal chemistry to radical generation and protection, and the structural basis for the transient interaction between NrdI and NrdF during radical assembly and reconstitution all represent open questions of significant mechanistic and biological interest. Understanding these processes at the level of conformational

dynamics, rather than through static structural snapshots alone, represents an important frontier in the structural biology of this enzyme class (Srinivas et al., 2018).

2.4 BioEmu-1: A Generative Deep Learning Framework for Protein Conformational Ensemble Generation

The study of protein conformational dynamics has historically been constrained by the significant computational cost associated with classical molecular dynamics (MD) simulations and by the technical limitations of experimental methods in capturing transient or low-population structural states. While advances in cryo-electron microscopy and time-resolved crystallography have expanded the accessible experimental conformational space, these approaches remain largely restricted to capturing the most populated or most stable conformational substates. Complementary computational approaches, including enhanced sampling MD, metadynamics, and replica exchange methods, have extended the accessible timescales and conformational diversity of simulation-based studies, yet remain computationally intensive and difficult to generalise across diverse protein systems.

A conceptually significant departure from these established methodologies was introduced by Microsoft Research with the development of BioEmu-1, a generative deep learning framework specifically designed to produce thermodynamically grounded protein conformational ensembles at substantially lower computational cost than conventional simulation approaches (Lewis et al., 2026). Unlike structure prediction models such as AlphaFold2, which are trained on curated databases of experimentally determined static structures and are optimised to predict the single most probable conformation for a given sequence, BioEmu-1 is trained on a large and heterogeneous dataset comprising over 200 milliseconds of molecular dynamics simulation trajectories across thousands of diverse protein systems, structural predictions for more than 100 million proteins, and over 500,000 experimental measurements of protein thermodynamic stability.

The architectural foundation of BioEmu-1 is a generative diffusion model that operates in the space of protein backbone conformations, using an AlphaFold2 Evoformer-based sequence encoder to provide evolutionary context and a score-based denoising network to learn the reverse mapping from noise to physically plausible protein structures. The model generates statistically independent structural samples from the equilibrium conformational ensemble of

a protein, with fine-tuning on MD simulation data weighted by Markov State Models (MSMs), ensuring that generated conformations are proportionally representative of their thermodynamic populations **(Lewis et al., 2026)**. Side-chain reconstruction is subsequently performed using HPacker, yielding all-atom structures suitable for downstream structural analysis.

Benchmarking comparisons have demonstrated that BioEmu-1 provides broader and more physically meaningful conformational sampling than ensemble approaches derived from AlphaFold2 through multiple sequence alignment (MSA) perturbation or dropout strategies, which generate structural diversity as a byproduct of sequence uncertainty rather than through direct learning of protein dynamics **(Lewis et al., 2026)**. BioEmu-1-generated ensembles exhibit improved coverage of conformationally distinct substates and of experimentally relevant functional transitions, while requiring less than a single GPU-hour for many protein systems, a computational efficiency gain of several orders of magnitude relative to conventional MD simulation of equivalent conformational space.

The integration of BioEmu-1 conformational ensembles with Markov State Model analysis provides a pathway to quantifying the kinetics and thermodynamics of protein conformational transitions, enabling the identification of metastable states, the estimation of inter-state transition rates, and the construction of free-energy landscapes from computationally generated ensemble data. Additionally, BioEmu-1 ensembles are compatible with cryo-EM-guided structural refinement pipelines such as CryoPhold, facilitating the integration of experimentally derived electron density information with computationally generated conformational diversity. These capabilities collectively position BioEmu-1 as a promising and broadly applicable tool for the structural investigation of dynamic enzyme systems such as Class Ib RNR, where transient conformational states associated with radical transfer, allosteric regulation, and catalytic gating have remained difficult to characterise through conventional structural methods alone.

MATERIAL AND METHODS

3.1 Molecular Dynamics Trajectory Collection and Preparation

Molecular dynamics (MD) trajectories were analysed using the MDAnalysis library. For each simulated system, topology files (.pdb format) and trajectory files (.xtc format) were sequentially loaded into the MDAnalysis environment. To maintain computational efficiency while preserving structural information, all analyses were restricted to protein C α (alpha-carbon) atoms.

Residue numbering was standardised using continuous sequence-based indexing; residues were renumbered sequentially from 1 to n, regardless of the original topology file numbering, to resolve non-canonical or discontinuous numbering arising from chain truncations, missing residues, or multi-domain systems.

This step was necessary because simulation structures may contain missing residues, chain breaks, insertions, or inconsistent numbering schemes. Standardised residue indexing ensured reproducible residue mapping across all analyses and enabled easier comparison with experimental protein structures and published datasets.

3.2 Structural Alignment and Global Motion Removal

To isolate intrinsic protein conformational dynamics from global translational and rotational artefacts, all trajectory frames were aligned to a reference structure before analysis. The reference frame was selected as the medoid conformation, the frame with the minimum summed pairwise RMSD to all other frames in the trajectory, thereby ensuring that the reference represented a central, representative conformation rather than a peripheral or outlying structure. Structural alignment was performed by least-squares fitting of C α coordinates using the `AtomGroup.align_to()` function in MDAnalysis, computing the optimal rotation matrix and centroid-based translation that minimised the sum of squared inter-atomic distances between the input and reference frames. Both rotational and translational transformations were applied, effectively removing global rigid-body degrees of freedom while preserving all internal conformational information.

3.3 Structural Stability Assessment: RMSD Analysis

Root-mean-square deviation (RMSD) was calculated for every trajectory frame relative to the medoid reference, yielding a time-series RMSD profile for each simulated system. RMSD profiles were visually inspected to delineate the equilibration phase, characterised by an initial rapid decrease in RMSD values as the system relaxed toward the equilibrium ensemble. The equilibration period (0-5 ns) was excluded from all subsequent analyses, retaining only the post-equilibration production phase. Convergence of the production phase was confirmed by the presence of a stable RMSD plateau, fluctuations within a narrow range (± 0.5 - 1.5 Å), and stationary mean and standard deviation values across sequential time windows. Mean RMSD and its standard deviation were computed over the production phase to quantify overall conformational stability.

3.4 Residue-Level Flexibility Analysis: RMSF

Residue-level conformational flexibility was characterised by calculating root-mean-square fluctuation (RMSF) for each residue across the production trajectory. RMSF was computed as the root-mean-square deviation of each residue's C α position from its time-averaged coordinate, providing a per-residue measure of thermal motion amplitude. Terminal residues were excluded from hotspot analyses to prevent intrinsic terminal mobility from obscuring functionally relevant flexibility within the protein core.

To identify statistically significant flexibility enrichments, raw RMSF values were z-score normalised across all analysed residues, yielding a standardised distribution with a mean of zero and unit standard deviation. Residues with z-scores above +1.0 were classified as displaying above-average flexibility, while those exceeding +1.5 were designated strong flexibility hotspots. A sliding-window moving average was subsequently applied to smooth local RMSF variation, and contiguous flexible regions were defined as segments where smoothed RMSF exceeded a z-score threshold of 1.0 for at least three consecutive residues.

3.5 Correlated Motion Analysis: Dynamical Cross-Correlation Matrix (DCCM)

Correlated inter-residue motions were characterised through construction of the dynamical cross-correlation matrix (DCCM). For each residue, displacement vectors were computed as

the deviation of the instantaneous C α position from the time-averaged coordinate. The cross-correlation coefficient between residues i and j was then calculated as the time-averaged dot product of their displacement vectors, normalised by the product of their respective root-mean-square displacements, yielding a dimensionless coefficient ranging from -1 (anti-correlation) to $+1$ (positive correlation). The resulting $n^{\text{residues}} \times n^{\text{residues}}$ DCCM matrix was visualised as a heatmap, with red indicating positive correlation, blue indicating negative correlation, and white indicating independent motion.

3.6 Residue Interaction Network Construction and Analysis

Residue interaction networks were constructed from the DCCM data using a graph-theoretical framework, with protein residues represented as nodes and dynamically correlated residue pairs represented as edges. Edges were defined between residue pairs exhibiting absolute cross-correlation coefficients exceeding a threshold of $\tau = 0.35$, corresponding approximately to the 90th percentile of the correlation distribution for uncorrelated residue pairs. Short-range sequential neighbours ($|i - j| \leq 2$) were excluded from network construction to enrich the network for long-range dynamic communication pathways. Network topological properties, including node degree, network density, and degree distribution, were computed using the NetworkX library in python.

3.7 Hub Identification and Centrality Analysis

Two centrality metrics were computed for all network nodes using the formula given below. Degree centrality was defined as the number of edges connected to each residue, normalised by $(N - 1)$, and reflected direct local connectivity.

$$C_D(i) = \text{Degree}_i = \sum_{j=1}^N \delta_{ij}$$

$$C_D^{\text{norm}}(i) = \frac{\text{Degree}_i}{N - 1}$$

Betweenness centrality quantified the fraction of shortest inter-residue paths that passed through each node, identifying residues that serve as critical bottlenecks for network-wide communication.

$$C_B^{norm}(i) = \frac{C_B(i)}{(N-1)(N-2)/2}$$

$$C_B^{norm}(i) = \frac{C_B(i)}{(N-1)(N-2)/2}$$

A composite hub score was derived as a weighted sum of normalised degree centrality (weight: 0.65) and normalised betweenness centrality (weight: 0.35), reflecting the greater importance of direct connectivity while accounting for the long-range routing function. Residues with hub scores in the top 15% of the distribution were designated hub residues.

$$\text{Hub Score}_i = 0.65 \times C_D^{norm}(i) + 0.35 \times C_B^{norm}(i)$$

Dynamic hub residues were subsequently identified by cross-referencing hub scores with RMSF data. Residues simultaneously displaying above-average flexibility ($z^{\text{RMSF}} \geq 1.0$) and high network centrality (top 15% hub score) were labelled as conformational switch residues. They were considered candidates for allosteric regulatory function, combining structural flexibility with network-level communication importance.

3.8 Quality Control and Reproducibility

Analytical reproducibility was ensured through fixed random seeds for all stochastic operations and detailed parameter logging. All the codes were version-controlled to allow reproducibility. The robustness of RMSF and DCCM results was assessed by repeating calculations on trajectory subsets, and network and hotspot analyses were validated across a range of correlation thresholds. Key flexibility and essential mode findings were cross-validated against alternative analytical implementations, and computed properties were compared with literature reference values where available. Statistical significance of RMSF comparisons was assessed using paired t-tests, with confidence intervals computed for correlation coefficients and hub scores evaluated by percentile ranking.

RESULTS

4.1 Conformational Dynamics of NrdB (PDB ID: 3EE4)

4.1.1 Global Structural Stability: RMSD Analysis

Structural stability of the BioEmu-generated NrdB ensemble was evaluated by computing the backbone RMSD of every sampled conformation relative to the medoid reference frame (frame 37). The mean RMSD across the ensemble was 2.59 Å, indicating that the generated conformers clustered reasonably close to the central representative structure and that the ensemble as a whole maintained overall fold integrity. The maximum RMSD of 8.19 Å, recorded at frame 6, reflected the existence of transiently expanded conformational states involving larger-scale structural displacements, while the minimum RMSD was effectively zero at the medoid frame itself (1.34×10^{-6} Å). The distribution of RMSD values confirmed that the BioEmu ensemble successfully captured both compact near-native conformations and more extended structural states consistent with the dynamic character expected of an R2-type radical-harboursing subunit.

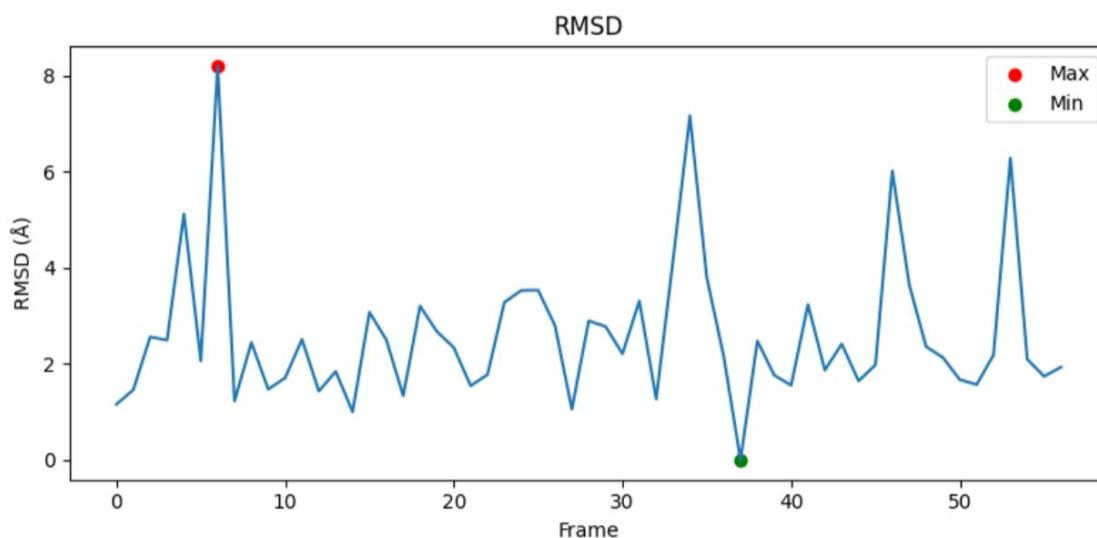


Figure 4 The backbone RMSD of every sampled conformation relative to the medoid reference frame (frame 37)

Table 1. Summary of RMSD statistics for the NrdB (3EE4) BioEmu ensemble.

Parameter	Value	Frame
Mean RMSD (Å)	2.59	All Frames
Maximum RMSD (Å)	8.19	Frame 6
Minimum RMSD (Å)	~0.00	Frame 37 (medoid)

4.1.2 Residue-Level Flexibility: RMSF Analysis

Residue-level flexibility was assessed through RMSF analysis, which revealed a heterogeneous distribution of conformational mobility across the NrdB sequence. The most flexible position in the ensemble was residue 1, with an RMSF of 19.48 Å, consistent with the intrinsic disorder of the unrestrained N-terminal tail. At the other extreme, residue 183 was the most structurally rigid, with an RMSF of only 0.52 Å, indicative of a well-anchored core region likely involved in maintaining the structural integrity of the metal-binding environment. Beyond these extremes, the RMSF profile identified two internally flexible regions, residues 75-89 and 142-155, which showed elevated fluctuations above the protein-wide mean following exclusion of terminal residues. These segments were designated candidate flexibility hotspots for subsequent network analysis.

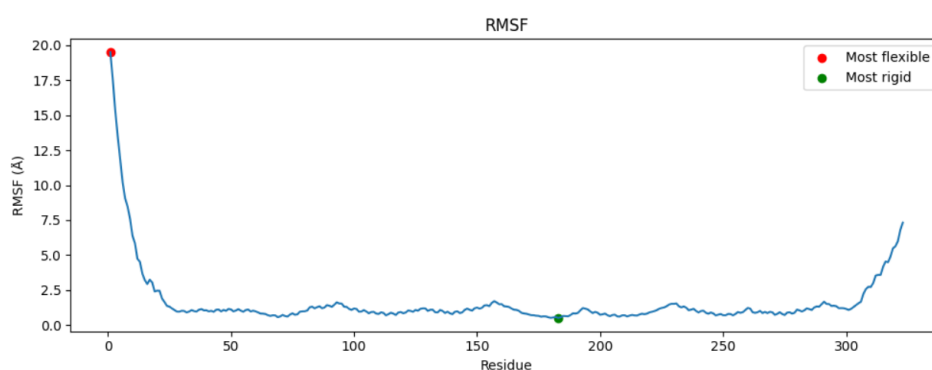


Figure 5. The RMSF analysis, which revealed a heterogeneous distribution of conformational mobility across the NrdB sequence

Table 2. Key flexibility statistics for the NrdB (3EE4) ensemble.

Residue	RMSF (Å)	Annotation
1	19.48	N-terminal tail (excluded from hotspot analysis)
183	0.52	Most rigid core residue
75-89	Elevated	Internal flexible region
142-155	Elevated	Hinge-like flexible region

4.1.3 Correlated Motions and Residue Interaction Network

Dynamic cross-correlation matrix (DCCM) analysis revealed the presence of significant long-range coordinated motions throughout the NrdB structure. The strongest positively correlated residue pair was 211-232 (DCCM = 0.97), indicating a high degree of directional synchrony between these two structurally distant segments. In contrast, the strongest negative correlation was observed between residues 134 and 166 (DCCM = -0.88), suggesting opposing domain-like displacements that may contribute to conformational regulation within the R2 scaffold.

Residue interaction network analysis identified a set of high-betweenness hub residues concentrated within the 103-151 range, particularly at positions 103, 112, 118, 132, 144, and 151. Many of these hubs were co-localised with the flexible regions identified by RMSF analysis, particularly the 75-89 and 142-155 segments, suggesting a structural coupling between conformational mobility and long-range communication efficiency. The region spanning residues 216-227 additionally showed moderate hub enrichment and overlapped substantially with the strongly correlated 211-232 segment identified in the DCCM, supporting the existence of a dynamically coordinated functional zone in this part of the protein. In contrast, the 278-290 region displayed comparatively limited hub connectivity.

Structural mapping of these network features onto the 3EE4 reference structure revealed a clear spatial organisation, with high-centrality hub residues (shown in red), flexibility hotspots (yellow), the metal-binding core environment (orange), and critical allosteric overlap residues (magenta spheres) forming a spatially coherent communication architecture within the NrdB fold (Figure 3)

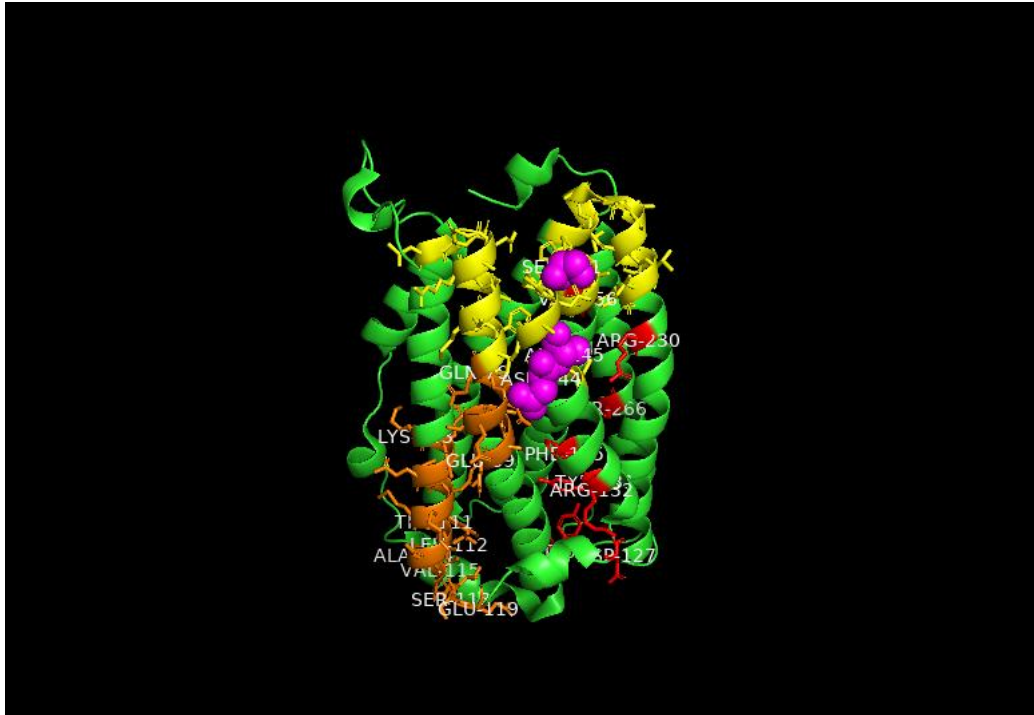


Figure 6. Structural visualisation of *NrdB* (PDB: 3EE4) mapped with dynamic and network features.

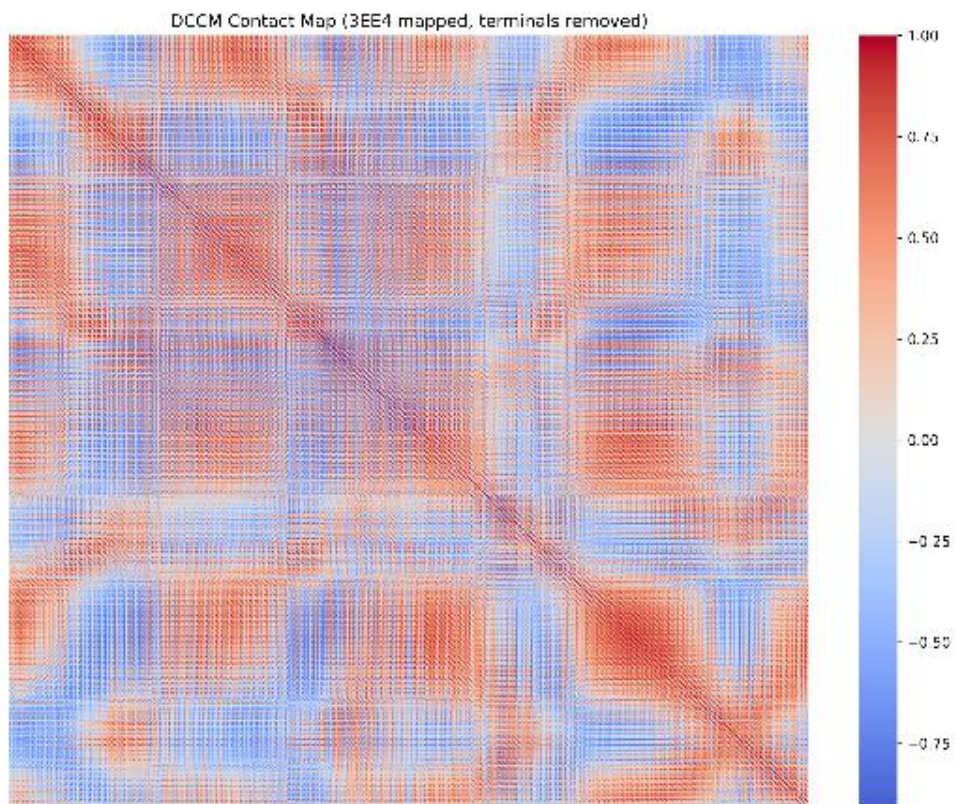


Figure 7. A Dynamic cross-correlation matrix (DCCM) contact map of *NrdB*

Table 3. Hub residue distribution across structural regions of NrdB (3EE4)

Region (Residues)	No. of Hub Residues	Classification
69-120 (Core Region)	9	Primary communication core
142-155 (Flexible Hinge)	4	Flexibility-communication overlap
216-230 (Surface / Loop)	2	Correlated functional zone
56-88 (N-terminal Core)	3	Peripheral communication nodes

4.1.4 Contact Map Analysis

Contact probability maps were computed from the aggregated BioEmu-augmented MD ensemble of NrdB to assess the preservation and modulation of inter-residue interactions relative to the reference crystal structure. Of the 289 residues successfully matched to the experimental structure across 57 analysed frames, 1,970 contacts were preserved between the crystal structure and the ensemble, confirming that the BioEmu workflow faithfully maintained the native contact topology of the R2 subunit. A total of 822 contacts present in the crystal structure were reduced or absent in the ensemble, concentrated in regions of elevated flexibility. Only 73 new contacts were gained in the ensemble, which were clustered in specific regions of the difference map, suggestive of transient loop rearrangements or channel-adjacent interactions not captured in the single static crystal structure.

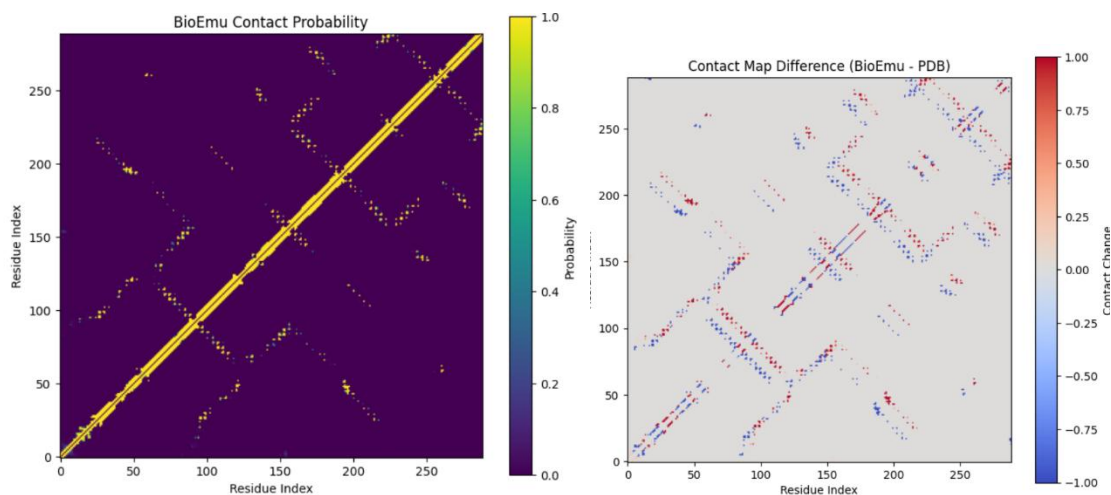


Figure 5: BioEmu contact probability map and contact difference analysis relative to the crystal structure.

Table 4. Inter-residue contact statistics for the NrdB BioEmu ensemble versus the 3EE4 crystal structure.

Contact Category	Count
Residues matched to the crystal structure	289
Frames analysed	57
Preserved contacts (crystal $\cap$ ensemble)	1,970
Lost contacts (crystal only)	822
Gained contacts (ensemble only)	73

4.2 Conformational Dynamics of NrdI from *Mycobacterium thermoresistibile* (PDB ID: 8J4V)

The BioEmu-generated conformational ensemble of NrdI (PDB: 8J4V), comprising 48 frames and 150 C α atoms, showed a mean RMSD of approximately 4.98 Å relative to the medoid reference frame (frame 15). The highest RMSD of 7.71 Å at frame 3 indicated the sampling of substantially displaced conformational substates, while the near-zero RMSD at the medoid confirmed that at least one ensemble member closely approximated the experimentally resolved structure. The comparatively high average RMSD relative to NrdB reflects the greater global conformational variability of NrdI, consistent with its role as a dynamic flavoprotein redox mediator. Crucially, subsequent flexibility analysis

demonstrated that this global variability was primarily attributable to disordered terminal regions rather than the FMN-binding catalytic core.

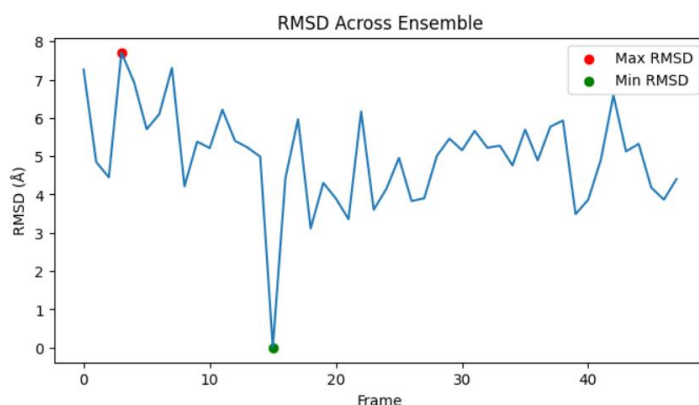


Figure 6. RMSD analysis across the ensemble for NrdI

Table 5. RMSD statistics for the NrdI (8J4V) BioEmu ensemble

Parameter	Value	Frame
Mean RMSD (Å)	~4.98	All Frames
Maximum RMSD (Å)	~7.71	Frame 3
Minimum RMSD (Å)	~0.0005	Frame 15 (medoid)
Frames analysed	48	--
C α atoms analysed	150	--

4.2.2 Residue-Level Flexibility: RMSF Analysis

RMSF analysis identified the C-terminal tail (residues 142-150) as the dominant source of conformational mobility in the NrdI ensemble, with residue 150 exhibiting the maximum fluctuation of 18.62 Å. This extreme terminal mobility is characteristic of solvent-exposed, structurally unconstrained termini and was excluded from the internal flexibility analysis. Following terminal exclusion, the major internal flexible hotspot was localised to residues 51-61, corresponding to the experimentally described 50_s-loop of NrdI. Within this segment, residue THR55 exhibited the highest internal RMSF, making it the dominant mobile position in the structured protein core. The adjacent FMN-binding contact residues, THR48, TYR49, GLY50, and GLY51, displayed substantially reduced flexibility, with RMSF values

well below the protein-wide mean, confirming the structural rigidity of the catalytic scaffold that positions the isoalloxazine ring of FMN.

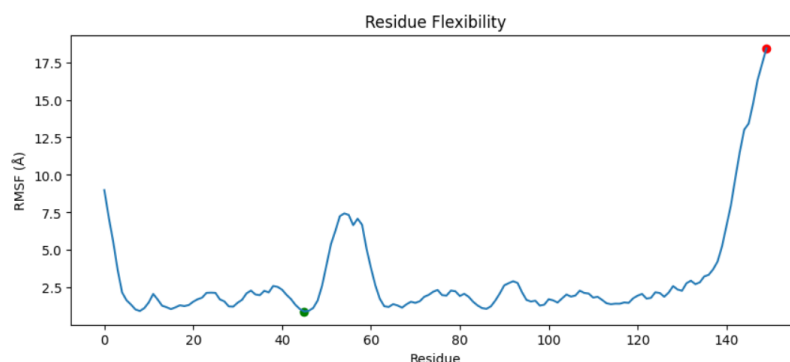


Figure 7. The RMSF analysis, which revealed a heterogeneous distribution of conformational mobility across the *NrdI* sequence

This functional separation between the rigid FMN-contact platform and the dynamic 50s-loop is visually apparent in the structural mapping of RMSF data onto the 8J4V reference frame, where the FMN-proximal residues (cyan) form a tightly constrained scaffold contrasting sharply with the highly mobile loop residues (red) extending away from the cofactor-binding site (Figure 7)

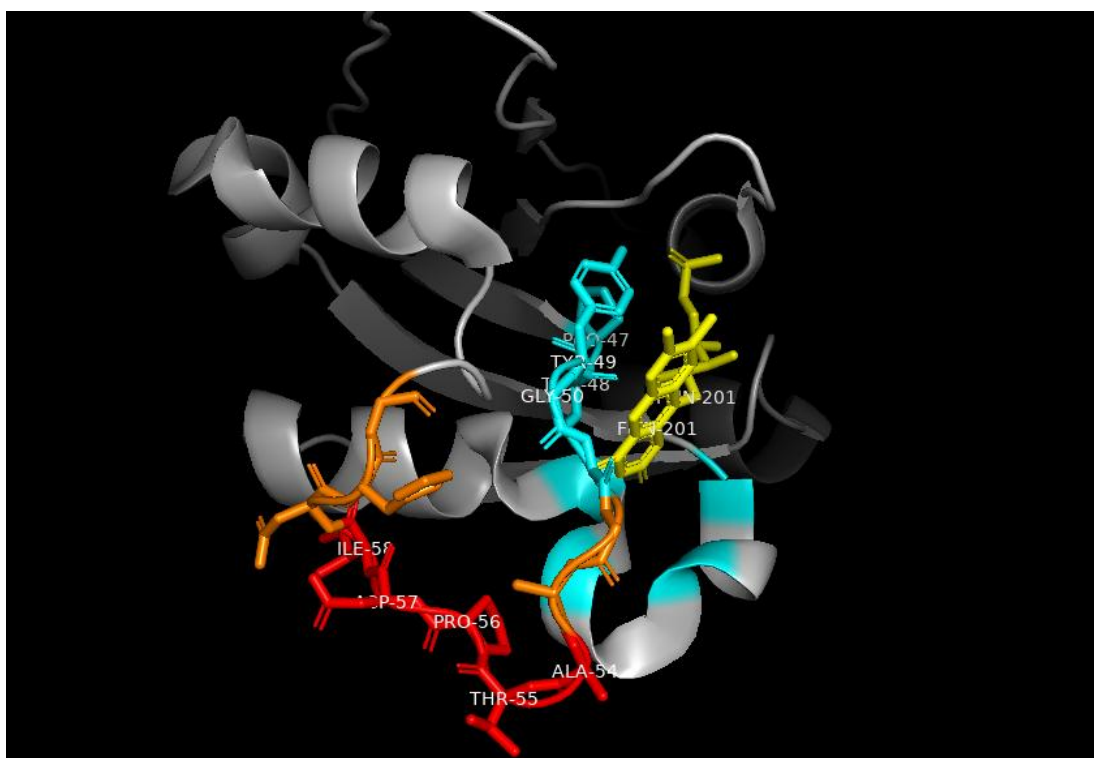


Figure 8. Structural visualisation of *NrdI* (PDB: 8J4V) showing the bipartite dynamic organisation.

Table 6. Key flexibility features of the NrdI (8J4V) ensemble.

Position / Region	RMSF (Å)	Annotation
Residue 150 (C-terminal)	~18.62	Terminal disorder (excluded)
Residues 51-61 (50s-loop)	Elevated (peak: THR55)	Primary internal hotspot
Residues 48-51 (FMN-contact)	Low (<global mean ~2.70)	Rigid catalytic scaffold

4.2.3 Correlated Motions and Communication Network

DCCM analysis, performed after removal of terminal residues and restricted to 99 core residues spanning positions 17-115, revealed a well-organised pattern of long-range correlated and negatively correlated motions intrinsic to the NrdI fold. The strongest positive correlation was observed between residues 43 and 82 ($r = 0.9582$), with several additional residue pairs in the 41-85 and 82-112 ranges also showing correlations above 0.94, indicating the existence of coherent dynamic communication pathways across the protein core. The strongest negative correlation was between residues 33 and 113 ($r = -0.5909$), with several neighbouring pairs displaying similarly negative values, suggesting compensatory opposing motions between structurally distant regions.

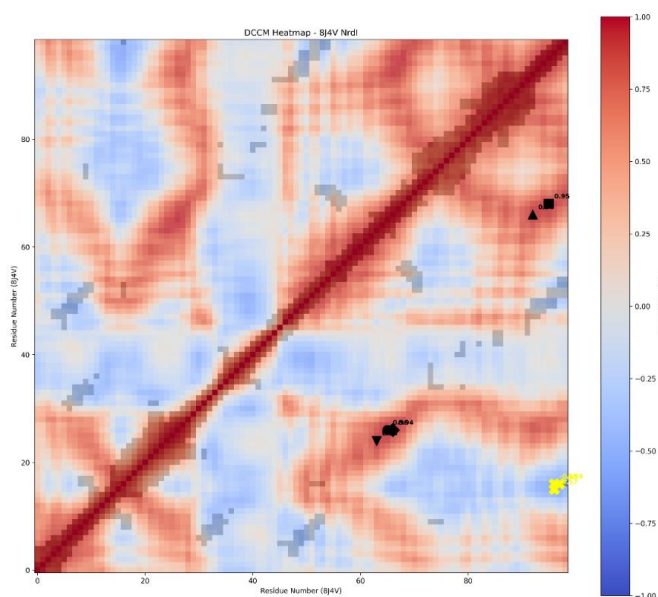


Figure 9. DCCM and Network Analysis of NrdI.

Betweenness centrality analysis identified PHE9 as the residue with the highest overall communication centrality, followed by **VAL68 and ARG114**. Within the core, **PRO47 and THR55** emerged as particularly notable hub residues. PRO47 is positioned at the structural junction between the rigid FMN-contact scaffold and the dynamic 50s-loop, while THR55 lies at the peak of flexibility within the loop itself. Together, these two residues define a structural communication bridge linking the catalytic core to the primary dynamic regulatory element of the protein.

Table 7. Key hub residues identified by betweenness centrality analysis in NrdI (8J4V)

Hub Residue	Betweenness Centrality	Structural Location
PHE9	Highest	N-terminal anchor
PRO47	High	FMN-core / 50s-loop junction
THR55	High	50s-loop peak (highest internal RMSF)
VAL68	Moderate-high	Core scaffold
ARG114	Moderate-high	Distal C-terminal region

4.3 Conformational Dynamics of the NrdF2:NrdI Complex from *Mycobacterium tuberculosis* (PDB ID: 8J4X)

4.3.1 NrdF2 Subunit - Global Stability and Flexibility

The BioEmu ensemble of the NrdF2 subunit from the 8J4X complex showed a mean RMSD of approximately 7 Å, reflecting substantial global conformational variability attributable largely to highly mobile terminal regions. Frame 33, the medoid reference, showed near-zero RMSD, confirming the structural centrality of this reference conformation. The highest RMSD was observed at frame 16, consistent with transient large-scale displacements. The most flexible residue in the raw ensemble was residue 324 at the C-terminus, while residue 67 was the most structurally rigid, indicative of a well-anchored core position.

Internal flexibility analysis, restricted to 242 structured core residues (positions 30-273) following terminal exclusion from a total of 284 sequence-mapped residues, revealed three localised flexibility hotspots. Residues 31-35 (N-terminal core), residues 88-98 (central loop region), and residues 210-218 (C-terminal proximal region). The highest internal RMSF

values were recorded at residues 90 (4.22 Å), 91 (4.18 Å), 89 (4.07 Å), 33 (4.05 Å), and 217 (4.00 Å), all substantially below the extreme terminal values and indicative of moderate but structurally meaningful mobility within the catalytic scaffold.

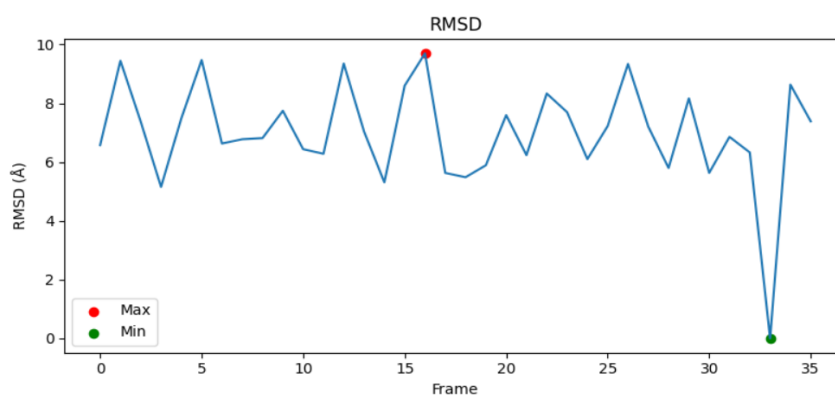


Figure 10. RMSD Analysis of NRDF2 (PDB ID:8J4X)

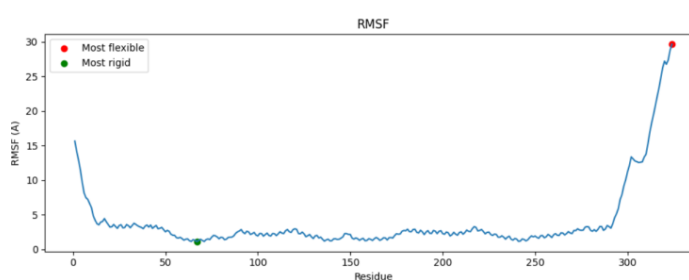


Figure 11. RMSF Analysis of NrdI

Table 8. Structural flexibility summary for NrdF2 (8J4X complex).

Parameter	Value	Notes
Mean RMSD (Å)	~7.00	Dominated by terminal flexibility
Residues mapped to 8J4X	284	—
Core residues analysed	242 (residues 30-273)	After terminal exclusion
Most flexible core region	Residues 88-98 (peak: residue 90, 4.22 Å)	Central loop hotspot
Most rigid residue	Residue 67	Stable core anchor

4.3.2 NrdF2 Communication Network

Network centrality analysis of the NrdF2 core identified two principal hub clusters. The most prominent was centred on residues 208-214, with residue 210 displaying the highest composite hub score (0.461), followed closely by residues 211, 209, and 208. A second major cluster was identified in the 88-98 region, encompassing residues 88, 93-98. Notably, these two hub clusters co-localised with the internal flexibility hotspots identified by RMSF analysis, suggesting that the structurally adaptive regions of NrdF2 are also the primary conduits for intra-protein signal propagation. The most prominent negatively correlated residue pairs in the DCCM were 55-216 ($r = -0.796$) and 125-272 ($r = -0.791$), indicating opposing long-range displacements across the NrdF2 core. The strongest positive correlation was observed between residues 88 and 207 ($r = 0.970$).

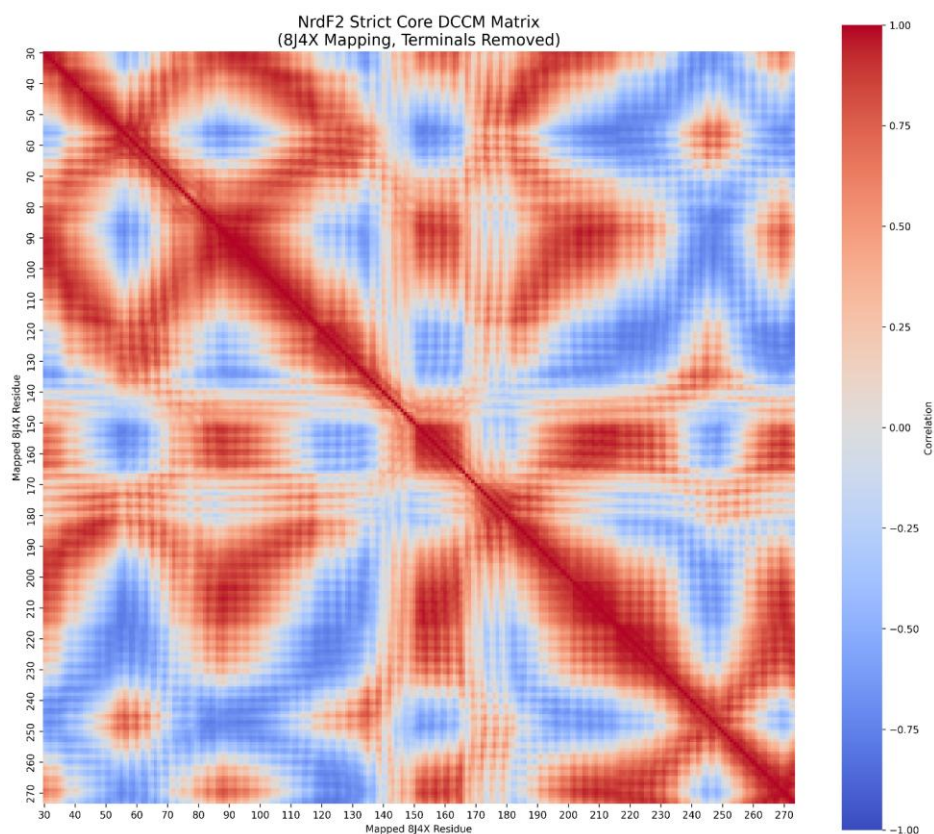


Figure 12. DCCM and Network Analysis of NRDF2(8J4X)



Figure 13. Structural overview of the NrdF2:NrdI complex (PDB: 8J4X). NrdF2 subunits are shown in grey and NrdI subunits in light blue.

The overall spatial distribution of hub residues across the NrdF2:NrdI complex is shown in Figure 12, where the NrdF2 catalytic subunit (grey) and NrdI accessory subunit (light blue) are annotated with their respective communication hubs (yellow spheres), the 50s-loop of NrdI (red), and the interfacial gating residues Glu97 and Arg30 (blue sticks), providing a system-level view of the communication architecture within the assembled complex

Table 9. Principal communication hub clusters identified in NrdF2 (8J4X complex)

Hub Cluster	Residues	Peak Hub Score
C-terminal cluster	208-214 (peak: residue 210)	0.461
Central cluster	88-98 (peak: residue 93-97)	Elevated

4.3.3 NrdI Subunit within the 8J4X Complex - Stability and Flexibility

The NrdI subunit from the 8J4X complex (51 frames, 150 Ca atoms) showed a mean RMSD of 4.51 Å relative to the medoid reference (frame 11), with a maximum of 7.85 Å at frame 17. RMSF analysis identified residue 149 as the most flexible position (17.31 Å), consistent with C-terminal disorder, while residue 47 was the most structurally rigid (0.89 Å). Following terminal exclusion, analysis of 97 core residues spanning positions 22-121 revealed the 50s-loop (residues 48-58) as the dominant internal flexibility hotspot, with the highest fluctuations at GLY51 (7.94 Å) and GLY50 (7.58 Å). The strong correspondence between the NrdI dynamics within the 8J4X complex and those of the homologous 8J4V

structure confirms the reproducibility of the 50_s-loop as the principal mobile element in NrdI across both mycobacterial systems.

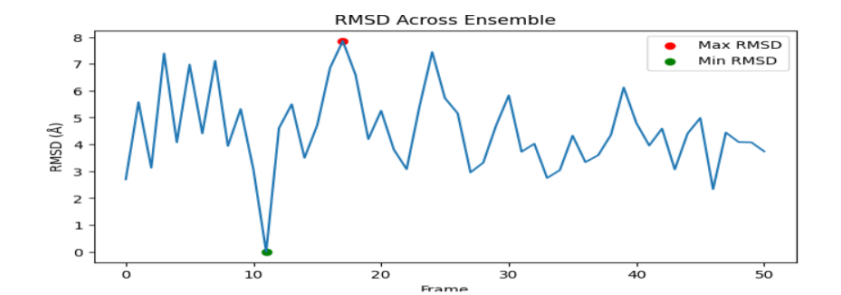


Figure 14. RMSD Analysis of NRDF2

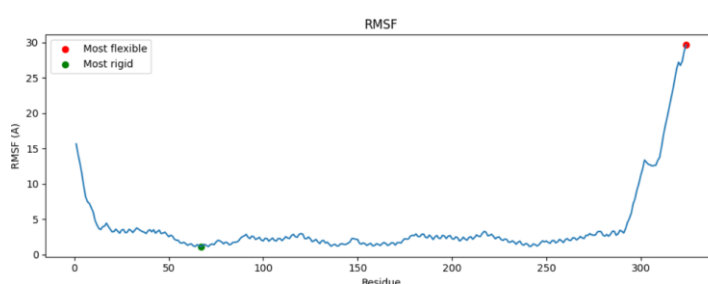


Figure 15. RMSF Analysis of NRDF2(PDB ID :8J4X)

4.3.4 NrdI Communication Network within the 8J4X Complex

DCCM analysis of the NrdI core in the 8J4X complex revealed strong long-range correlated motions consistent with those observed in the 8J4V analysis. The strongest positive correlation was between residues 43 and 82 ($r = 0.961$), with residue pairs 44-82 and 45-84 also exceeding 0.95. Anti-correlated motions were concentrated between residues 49 and 57 ($r = -0.613$) and neighbouring pairs, indicating localised opposing displacements within the 50_s-loop region itself.

Betweenness centrality analysis identified residue 54 as the top hub within the NrdI network (centrality: 0.1468), followed by residues 75 (0.1358), 95 (0.1172), 92 (0.0889), and 31 (0.0760). The positioning of residue 54 at the intersection of the 50_s-loop and the broader NrdI scaffold confirmed its role as the primary intramolecular communication node, linking the most mobile region of the protein to the wider conformational ensemble.

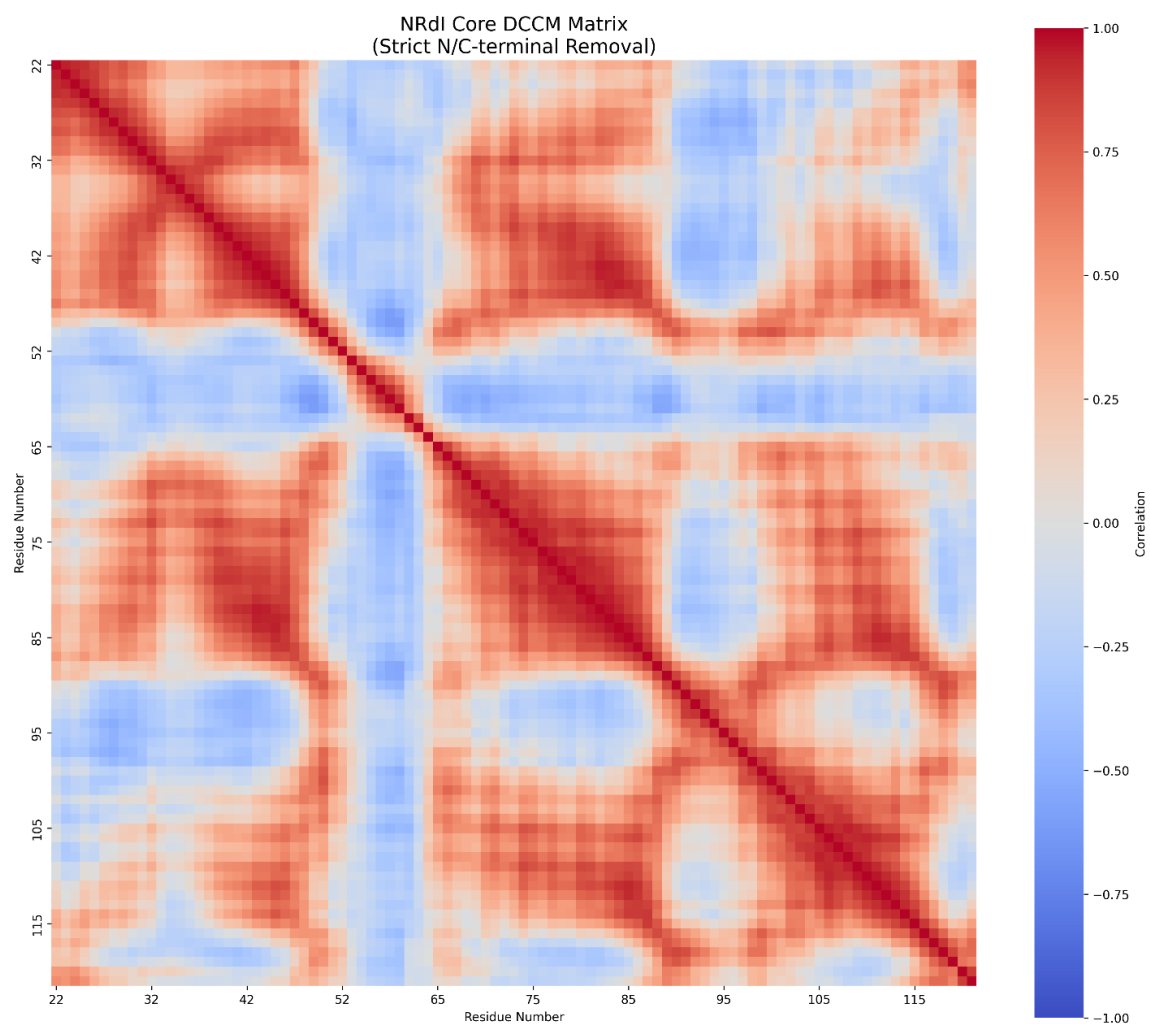


Figure 16.. DCCM and Network Analysis of NRdI from the complex

Table 10. Hub residue centrality values for NrdI in the 8J4X complex.

Hub Residue	Betweenness Centrality	Structural Location
Residue 54	0.1468 (highest)	50s-loop / scaffold junction
Residue 75	0.1358	Core scaffold
Residue 95	0.1172	Proximal to Glu97 gating region
Residue 92	0.0889	Core scaffold
Residue 31	0.0760	N-terminal anchor

DISCUSSION

9.1 BioEmu Captures Functionally Relevant Conformational Landscapes

The conformational ensembles generated by BioEmu for all three protein systems, NrdB, NrdI (8J4V), and the NrdF2:NrdI complex (8J4X) demonstrated a consistent structural pattern, a stable, well-conserved catalytic core coexisting with discrete regions of elevated conformational mobility. This organisation, in which structural rigidity and dynamic flexibility are spatially segregated into adjacent but non-overlapping modules, is a hallmark of enzymes that must simultaneously maintain precise active-site geometry and accommodate the conformational rearrangements required for catalysis. The BioEmu ensembles correctly reproduced this organisation across all three systems, validating the generative approach as a biologically meaningful sampling strategy for proteins of this class.

For **NrdB**, the mean RMSD of 2.59 Å indicated that the ensemble sampled conformational space in the vicinity of the native fold, while the upper RMSD tail (up to 8.19 Å) captured transient expanded states that would be inaccessible or rare in short conventional MD simulations. The contact map analysis provided further validation, 1,970 preserved contacts confirmed that the native topology of the R2 subunit was maintained across the ensemble, while the 822 lost and 73 gained contacts reflected genuine dynamic of the protein rather than non-physical structural disruption. Together, these observations support the conclusion that the BioEmu-generated ensemble represents a physically realistic and structurally grounded model of NrdB conformational space.

The **Mn/Fe** R2-like oxidase scaffold exhibits a structurally defined **V71–Y162** crosslink region that is central to cofactor-dependent oxidative maturation (Lebrette et al., 2009). Flexible segments around residues 144-155 and 216-227 likely contribute to conformational gating and structural adaptation of the ligand-access channel. Network hubs identified near the 70-120 region are structurally coupled to the metal-binding core and may indirectly stabilise the reactive centre, although direct roles in electron transfer are not explicitly established by crystallographic evidence

5.2 The 50s-Loop of NrdI is the Principal Dynamic Regulatory Element

Across both NrdI systems examined, the isolated 8J4V structure from *M. thermoresistibile* and the NrdI component of the 8J4X complex from *M. tuberculosis*, the 50s-loop

consistently emerged as the dominant internal flexible hotspot. The convergence of this finding across two independently analysed systems with 86% sequence identity provides strong evidence that the 50s-loop is a conserved structural feature of NrdI that is intrinsically predisposed to conformational mobility rather than being dynamically stabilised by crystal packing or ligand contacts in the experimental structures.

This finding is directly interpretable in light of the experimentally established function of the 50s-loop. The crystallographic study of the 8J4X complex identified the 50s-loop as the structural element that undergoes redox-state-dependent conformational switching in the oxidised complex, the loop adopts a conformation compatible with the Glu97(NrdI)-Arg30(NrdF2) ionic contact that partially constricts the oxygen-access tunnel, while in the reduced state, the loop repositions such that the carbonyl oxygen of Gly52 can directly hydrogen-bond to the N5 atom of FMN, facilitating electron transfer and tunnel opening. The pre-existing flexibility of this loop, as captured in the BioEmu ensemble, is consistent with a conformational selection mechanism in which the protein samples conformations compatible with the reduced-state geometry even in the absence of the redox trigger.

The identification of GLY50 and GLY51 as the most mobile positions within the 50s-loop is also important to note. Glycine residues impose minimal backbone dihedral restrictions and are commonly found at points of maximum conformational flexibility in functional loops. Their presence at the apex of the 50s-loop is likely a sequence-encoded feature that confers the high-amplitude motion required for efficient tunnelling and gating, and their high RMSF values across both NrdI systems confirm that this flexibility is an intrinsic thermodynamic property of the loop rather than a simulation artefact.

9.2 Residue 54 and PRO47 Define a Communication Bridge in NrdI

Network centrality analysis consistently identified residue 54 and PRO47 as the principal intramolecular communication hubs in NrdI. These two residues are spatially adjacent, with PRO47 lying at the junction between the rigid FMN-binding platform and the mobile 50s-loop, and residue 54 positioned within the peak flexibility region of the loop itself. Their simultaneous identification as high-betweenness residues or residues through which a disproportionate number of shortest inter-residue communication paths pass indicates that they collectively constitute a structural bridge linking the dynamic regulatory element of NrdI to the remainder of the protein fold.

This structural positioning has a straightforward mechanistic implication. PRO47, as a proline residue, imposes a fixed backbone dihedral at the loop-scaffold boundary, providing a rigid anchoring point from which the 50_s-loop can pivot during redox-driven conformational transitions. Residue 54, in contrast, lies within the mobile region and acts as an amplifier and distributor of loop motion into the broader NrdI network. Together, they define an architectural unit that is well-suited to converting chemical (redox) input at the FMN cofactor into the mechanical output required for tunnel gating and NrdF2 interaction. While this interpretation is inferred from the conformational ensemble rather than directly demonstrated by perturbation experiments, it is fully consistent with the experimental observations reported for the 8J4X crystal structure and represents a testable mechanistic hypothesis for future mutagenesis studies.

5.4 NrdF2 Maintains a Stable Catalytic Core with Localised Communication Hotspots

In contrast to the pronounced loop dynamics observed in NrdI, NrdF2 displayed a more restrained and evenly distributed flexibility profile, consistent with its role as a catalytic β -subunit that must preserve precise di-manganese coordination geometry for radical generation. The most flexible internal regions, residues 88-98 and 210-218, showed RMSF values in the 4.0-4.2 Å range, substantially lower than the dominant 50_s-loop motion in NrdI, confirming that NrdF2 operates within a narrower conformational envelope.

The two principal communication hub clusters identified in NrdF2, centred on residues 88-98 and 208-214, show spatial proximity to the metal-coordinating residues of the active site. The central cluster (88-98) lies in the structural neighbourhood of Glu103 and His106, both directly involved in Mn coordination, while the C-terminal cluster (208-214) is adjacent to the His200 and Glu197 ligands. Although none of the hub residues are themselves metal ligands, their positions as second-shell communication nodes suggests that they may participate in transmitting conformational signals between the protein surface and the catalytic core. This second-shell communication architecture, in which structural information propagates through a network of well-connected residues rather than through direct active-site contacts, is consistent with the general paradigm of allosteric regulation in metalloenzymes.

5.5 Asymmetric Functional Organisation of the NrdF2:NrdI Complex

Considered together, the dynamics of NrdF2 and NrdI within the 8J4X complex reveal a pronounced functional asymmetry. NrdI contributes the primary dynamic element of the assembly organised around a restricted set of high-centrality hub residues that link loop motion to the broader protein scaffold. NrdF2, in contrast, provides the catalytic stability required for metallocofactor chemistry, its limited internal flexibility appearing to serve a fine-tuning rather than a gating role. This contrast is consistent with the experimentally proposed division of labour in the Class Ib RNR system, in which NrdI acts as the conformationally adaptable redox mediator and NrdF2 provides the persistent structural environment for radical initiation.

At the subunit interface, the experimentally characterised Glu97(NrdI)-Arg30(NrdF2) ionic interaction provides the structural link between these two functional regimes. The identification of residue 95 as a moderately ranked communication hub in NrdI suggests that the NrdI communication network extends toward this interfacial gating residue, potentially coupling intramolecular 50s-loop motion to intermolecular interaction dynamics. While this connection is inferred from network topology rather than directly demonstrated experimentally, it offers a plausible mechanistic basis for the redox-sensitive modulation of the subunit interface observed in the crystal structures.

CONCLUSION & FUTURE SCOPE

Ribonucleotide reductase has long been recognised as an enzymologically remarkable system, yet the mechanistic understanding of its function has remained constrained by the inherent limitations of static structural methods. While crystallographic and cryo-electron microscopy studies have established the foundational architecture of the Class Ib NrdE-NrdF-NrdI system and identified key residues involved in radical generation, allosteric regulation, and inter-subunit communication, the transient conformational states that govern these processes have remained largely inaccessible to experimental characterisation alone. The present study addressed this gap by applying BioEmu, a generative diffusion model trained on large-scale molecular dynamics data, to generate thermodynamically grounded conformational ensembles for NrdB, NrdI, and the NrdF2:NrdI complex, and by integrating these ensembles with RMSF-based flexibility profiling, dynamical cross-correlation analysis, and residue interaction network characterisation. Together, these analyses revealed a consistent dynamic organisation across all three systems, a structurally rigid catalytic core preserving the geometry required for metallocofactor chemistry and radical initiation, coexisting with discrete, functionally significant regions of elevated conformational mobility. The 50_s-loop of NrdI emerged as the principal dynamic regulatory element of the complex, with its intrinsic pre-sampled flexibility providing a thermodynamic basis for redox-coupled tunnel gating through the conserved Glu97-Arg30 interfacial interaction. Network centrality analysis further identified PRO47 and residue 54 in NrdI, and hub clusters at residues 88-98 and 208-214 in NrdF2, as the primary nodes of intramolecular communication, suggesting that conformational signals are propagated through a restricted and structurally strategic set of residues rather than being distributed uniformly across the fold.

These findings open several important scope of study for future. The conformational ensembles generated here were produced for individual subunits rather than the fully assembled NrdF2:NrdI complex, and extending BioEmu-based sampling to the explicit heterocomplex, incorporating the FMN cofactor, di-manganese centre, and interfacial contacts, would enable direct interrogation of inter-subunit communication and redox-dependent conformational coupling in a more complete structural context. Integration of the generated ensembles with cryo-EM density data through frameworks such as CryoPhold would provide an additional layer of experimental validation and allow

ensemble refinement. Furthermore, the mechanistic hypotheses generated by network analysis especially, the proposed roles of PRO47, residue 54, and the NrdF2 hub clusters, represent concrete targets for site-directed mutagenesis and biophysical validation. Beyond this specific system, the workflow established here shows that BioEmu-driven ensemble generation combined with other graph-based analysis forms a broadly applicable and computationally efficient framework for investigating conformational regulation in radical enzymes and metalloprotein complexes. Its extension to other RNR classes, inhibitor-bound states, and disease-associated variants represents a natural and scientifically productive direction for continued research.

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APPENDIX A

Software Tools, Databases, and Computational Resources

The following software packages, databases were used in this study. All tools are freely available for academic use unless otherwise stated.

A.1 Core Computational Tools

Tool / Package	Version	Purpose	Reference / URL
BioEmu-1	v1.0	Conformational ensemble generation	Lewis et al., 2026 / GitHub
MDAnalysis	2.7.0	MD trajectory parsing and analysis	Michaud-Agrawal et al., 2011
NetworkX	3.3	Graph construction and centrality analysis	Hagberg et al., 2008
NumPy	1.26	Numerical computation	Harris et al., 2020
Matplotlib	3.8	Data visualisation and plotting	Hunter, 2007
HPacker	—	Side-chain reconstruction from BioEmu ensembles	BioEmu pipeline
PyMOL	2.5	Molecular structure visualisation	Schrödinger LLC
Python	3.10	Primary scripting language	python.org

A.2 Databases and Structure Repositories

Resource	Content Used	Accession / URL
RCSB Protein Data Bank (PDB)	NrdB (3EE4), NrdI (8J4V), NrdF2:NrdI (8J4X)	rcsb.org
UniProtKB / Swiss-Prot	Sequence verification for RNR subunits	uniprot.org
AlphaFold Protein Structure Database	Pre-trained structural priors for BioEmu	alphafold.ebi.ac.uk